0.00011001100110011001100110011 (the exact value of the representation 0.00011001100110011001100110011 of 1/10 in the 24-fixed point register however, is 209715/2097152) . As we see, this is not an exact representation of 1/10. It would take infinite numbers of bits to represent 1/10 exactly. So, the error in the representation is (1/10-209715/2097152) which is approximately 9.5E-8 seconds.

On the day of the attack, the battery on the Patriot missile was functioning for 100 consecutive hours, hence causing an error of 9.5E-8x10x60x60x100=0.34 seconds (10 clock cycles in a second, 60 seconds in a minute, 60 minutes in an hour).

The shift of the defence gates calculated due to the error of 0.342 seconds was calculated as 687m. For the Patriot missile defence system, the target is considered out of range if the shift is more than 137m. Hence the incoming missile was considered out of range and was not targeted resulting in the death of 28 Americans in the barracks of Saudi Arabia.

NORAD reports U.S. was under missile attack(1980):

On June 3, 1980, at about two-thirty in the morning, computers at the National Military Command Center, at the headquarters of the North American Air Defense Command (NORAD) issued an urgent warning: the Soviet Union had just launched a nuclear attack on the United States.

Software could have ended the cold war in a rather catastrophic manner